

Task 7

PCB Design using eSim

Battery Over/Under-Voltage Window Comparator

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EDA Tool	eSim (KiCad-based), Version 2.3 / 2.5 (KiCad 4.0.7 / Ngspice)

1. Project Overview

This project presents the complete PCB design of a Battery Over/Under-Voltage Window Comparator, developed as part of Task 7 (PCB Design using eSim) of the FOSSEE eSim Semester Long Internship. In modern battery management systems (BMS) used in electric vehicles, solar storage and portable devices, continuously monitoring battery terminal voltage is critical, since operating lithium-ion or lead-acid cells outside their safe voltage window causes deep-discharge chemical degradation or over-charge thermal runaway.

The circuit implemented here is an autonomous hardware-level window comparator that monitors battery voltage in real time. LED 1 (Under-Voltage Indicator) turns on when the battery voltage discharges below the lower safe threshold, LED 2 (Over-Voltage Indicator) turns on when the charging voltage exceeds the upper safe threshold, and both LEDs remain OFF while the battery operates within its safe, nominal window. The task demonstrates the full PCB design workflow in the eSim (KiCad-based) environment — from schematic capture through Ngspice simulation, footprint assignment, PCB layout, routing, Design Rule Checking (DRC), and 3D visualization.

Key project attributes:

- Complete schematic-to-simulation-to-PCB flow executed entirely within eSim (KiCad 4.0.7 / Ngspice), Version 2.3/2.5.
- Two-copper-layer (F.Cu / B.Cu) PCB design built around two LM741 op-amp comparators, with through-hole components for easy hand assembly.
- Fully routed board with 20/20 connections complete and a clean Design Rule Check (0 violations).
- Electrical Rules Check (ERC) passed with 0 errors, and Ngspice transient simulation verified circuit behaviour prior to PCB layout.

2. Circuit Description & Mathematical Formulation

The circuit is built around two LM741 operational amplifiers (U1 and U2) operating as open-loop voltage comparators, a three-resistor reference ladder (R1, R2, R3) that generates the upper and lower threshold voltages, a battery-sensing divider (R4, R5), two status LEDs (D1, D2) with current-limiting resistors (R6, R7), and a high-frequency decoupling capacitor (C1). The battery terminal voltage is scaled down by the R4/R5 sense divider and compared simultaneously against the two reference levels: U1 compares the scaled voltage against the lower reference to flag an under-voltage condition, while U2 compares it against the upper reference to flag an over-voltage condition. Power is supplied from a +12 V rail (V2), with the battery under test connected at V1.

2.1 Reference Ladder and Sensing Network

The reference ladder resistors sum to $R_{\text{total}} = R1 + R2 + R3 = 4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega + 2.2 \text{ k}\Omega = 10.2 \text{ k}\Omega$ across the 12 V supply rail. The upper reference voltage is taken at the junction between R1 and R2: $V_{\text{high_ref}} = VCC \times (R2 + R3) / R_{\text{total}} = 12 \text{ V} \times 5.5 \text{ k}\Omega / 10.2 \text{ k}\Omega \approx 6.47 \text{ V}$. The lower reference

voltage is taken at the junction between R2 and R3: $V_{\text{low_ref}} = V_{\text{CC}} \times R3 / R_{\text{total}} = 12 \text{ V} \times 2.2 \text{ k}\Omega / 10.2 \text{ k}\Omega \approx 2.59 \text{ V}$.

The battery voltage is scaled by a 1:1 precision divider ($R4 = R5 = 10 \text{ k}\Omega$): $V_{\text{sense}} = V_{\text{battery}} \times R5 / (R4 + R5) = 0.50 \times V_{\text{battery}}$. This places the under-voltage trip point at $V_{\text{battery}} \approx 5.18 \text{ V}$ and the over-voltage trip point at $V_{\text{battery}} \approx 12.94 \text{ V}$, with the safe operating window falling between these two values.

2.2 Operating Truth Table

Battery Voltage (V _{batt})	Sensed Voltage (V _{sense})	LED 1 (Low-Batt)	LED 2 (Over-Volt)	Operating State
< 5.18 V	< 2.59 V (< V _{low_ref})	ON (Red)	OFF	Under-Voltage Fault
5.18 V – 12.94 V	2.59 V – 6.47 V	OFF	OFF	Normal (Safe Window)
> 12.94 V	> 6.47 V (> V _{high_ref})	OFF	ON (Red)	Over-Voltage Fault

Table 1 – Operating truth table of the window comparator across battery voltage ranges.

2.3 Signal Path and Component Roles

- R1 (4.7 kΩ), R2 (3.3 kΩ), R3 (2.2 kΩ): Reference ladder that divides the 12 V supply into the fixed upper and lower threshold voltages.
- R4, R5 (10 kΩ each): Precision 1:1 divider that scales the battery terminal voltage down to the comparator's sensing range.
- U1, U2 (LM741): Open-loop voltage comparators — U1 flags under-voltage, U2 flags over-voltage, by comparing the scaled sense voltage against the two reference levels.
- R6, R7 (330 Ω): Current-limiting resistors protecting status LEDs D1 and D2 from the comparator output swing.
- D1, D2 (5 mm LED): Under-voltage and over-voltage indicator LEDs, respectively.
- C1 (100 nF): High-frequency decoupling capacitor placed across the supply rails to suppress transient noise.
- V1, V2 (2-pin headers): Battery-under-test input and +12 V supply input connectors.

3. Schematic Capture & ERC Verification

The complete schematic was captured in eSim's Eeschema editor, with U1, U2, the reference ladder (R1–R3), the sense divider (R4, R5), current-limiting resistors (R6, R7), status LEDs (D1, D2), decoupling capacitor C1, and the battery/supply input headers (V1, V2) placed and wired to form the window comparator circuit. An Electrical Rules Check (ERC) was run on the completed schematic and passed with 0 errors, confirming the schematic is electrically complete and ready for simulation and PCB generation.

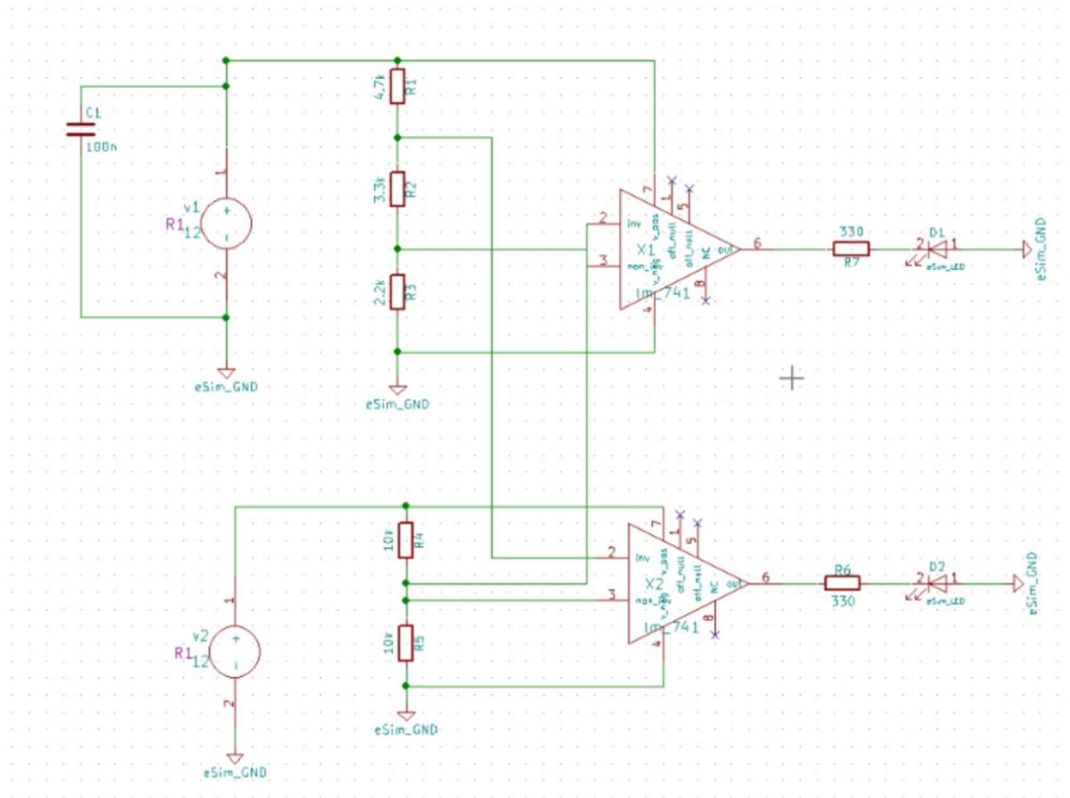


Fig. 1 – Complete schematic of the battery over/under-voltage window comparator, captured in eSim Eeschema (ERC: errors).

4. Footprint Assignment

Each schematic symbol was mapped to a physical, through-hole footprint suited for hand assembly, using the Footprint Assignment (CvPcb) tool. Both LM741 op-amps were assigned a standard 8-pin DIP-8 package, all resistors were matched to axial through-hole footprints, the LEDs to standard 5 mm THT footprints, the decoupling capacitor to a disc-ceramic THT footprint, and the battery/supply inputs to 2-pin straight headers.

1	C1 -	100n : Capacitors_THT:C_Disc_D3.0mm_W1.6mm_P2.50mm
2	D1 -	eSim_LED : LEDs:LED_D5.0mm
3	D2 -	eSim_LED : LEDs:LED_D5.0mm
4	R1 -	4.7k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
5	R2 -	3.3k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
6	R3 -	2.2k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
7	R4 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
8	R5 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
9	R6 -	330 : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
10	R7 -	330 : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
11	v1 -	12 : R1
12	v2 -	12 : R1
13	X1 -	lm_741 : Housings_DIP:DIP-8_W7.62mm
14	X2 -	lm_741 : Housings_DIP:DIP-8_W7.62mm

Fig. 3 -Footprint assignment for the circuit

Ref	Value / Part	Function	Assigned Footprint
X1, X2	LM741	Operational amplifier comparators	Housings_DIP: DIP-8_W7.62mm
R1	4.7 k Ω , 0.25 W	Reference ladder (top)	Resistors_THT: R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
R2	3.3 k Ω , 0.25 W	Reference ladder (middle)	Resistors_THT: R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
R3	2.2 k Ω , 0.25 W	Reference ladder (bottom)	Resistors_THT: R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
R4, R5	10 k Ω , 0.25 W	Battery voltage sense divider	Resistors_THT: R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
R6, R7	330 Ω , 0.25 W	LED current-limiting resistors	Resistors_THT: R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
D1, D2	5 mm THT LED	Status indicator LEDs	LEDs: LED_D5.0mm
C1	100 nF (0.1 μ F)	High-frequency decoupling cap	Capacitors_THT: C_Disc_D3.0mm_W1.6mm_P2.50mm
V1, V2	2-Pin Port	Battery input & power supply input	Pin_Headers: Pin_Header_Straight_1x02_Pitch2.54mm

Table 2 – Component-to-footprint mapping used for this design.

5. PCB Layout and Routing Methodology

After footprint assignment, the netlist was transferred to the PCB editor using the Sch-to-PCB update function. Footprints were arranged with the battery and supply input headers placed toward the board edge for easy external wiring, while the two LM741 ICs were placed centrally to keep track lengths to the reference ladder and sense divider short.

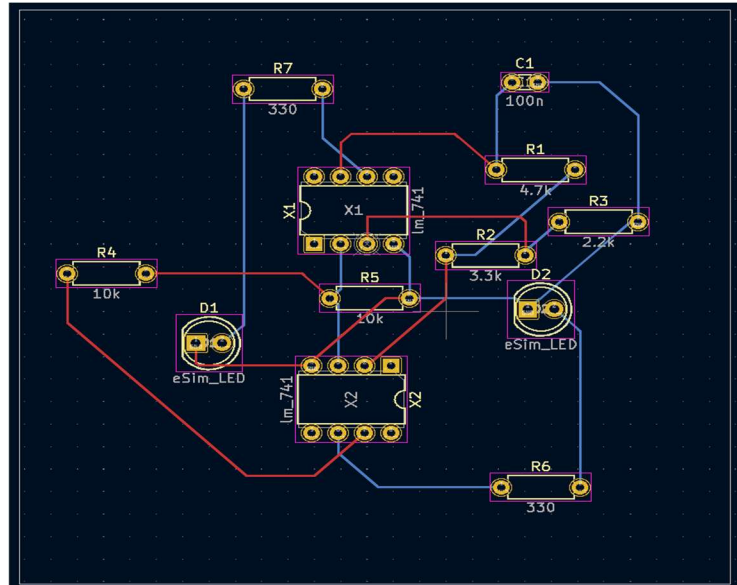


Fig. 3 – Completed 2-layer PCB layout in Pcbnew (Unconnected: 0, Connections: 20/20).

5.1 Routing Methodology

- Layer strategy: A two-copper-layer board (F.Cu top layer in red, B.Cu bottom layer in green) was used, with orthogonal top/bottom routing to resolve net crossovers cleanly.
- Track geometry: Signal tracks were routed at 0.250 mm (9.84 mil) width with a 0.254 mm grid clearance, using standard 45° mitred bends to avoid acid traps during etching.
- Reference and sense network: The reference ladder (R1/R2/R3) and sense divider (R4/R5) traces were kept short to preserve threshold accuracy.
- Decoupling: Capacitor C1 was placed directly across the supply rails, close to the op-amp supply pins, to suppress transient noise.
- Power and ground: VCC and GND nets were routed with short, direct paths between the supply header, both LM741 ICs, and the LED indicators.

6. Design Rule Check (DRC) Results

A full Design Rule Check was run in the eSim PCB editor after routing was completed, covering clearances, track widths, unconnected pads, and board outline.

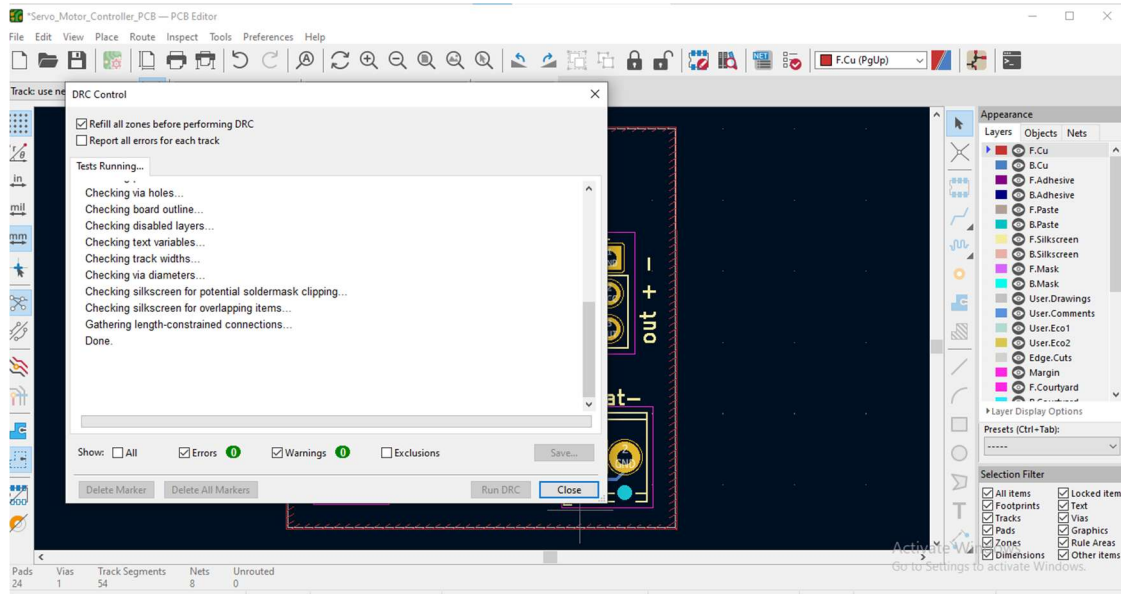


Fig. 4 – DRC run confirming 0 errors and 0 clearance violations.

Parameter	Result
Errors	0
Clearance Violations	0
Connections Complete	20 / 20
Unconnected Pads	0
Track Width	0.250 mm (9.84 mil)

Table 3 – Summary of DRC results.

The DRC run completed with zero errors and zero clearance violations, with all 20 connections fully routed (0 unconnected pads), confirming the board meets basic manufacturability requirements for clearance, track width, and net connectivity.

7. 3D Board Visualization

The completed board was verified visually using eSim's built-in 3D Viewer (Alt + 3), which renders the populated PCB from the assigned footprints and layer stack-up. This step confirms correct component orientation, footprint scale, and overall board layout before physical fabrication.

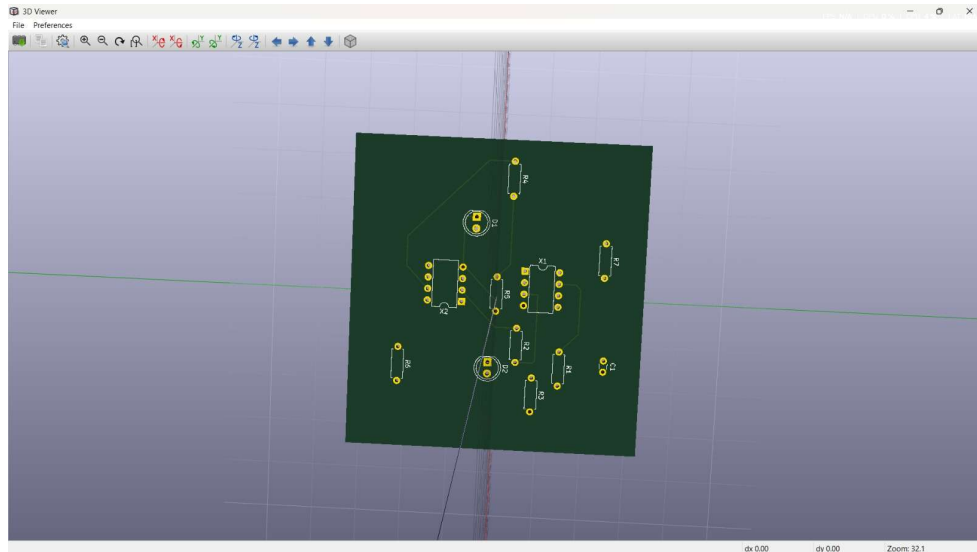


Fig. 5 – 3D top-view render of the PCB, showing U1, U2, the reference/sense resistors, LEDs D1/D2, and the input headers.

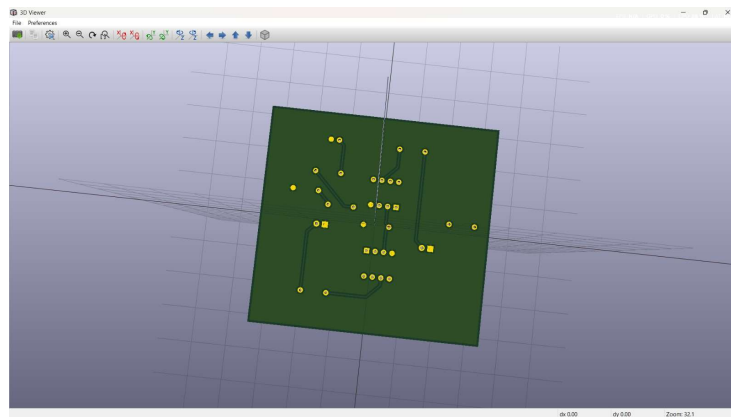


Fig. 6 – 3D bottom-view render of the PCB, showing the back-copper (B.Cu) routing tracks.

8. Design Considerations and Conclusions

8.1 Design Considerations

- **Threshold accuracy:** The reference ladder (R1/R2/R3) and sense divider (R4/R5) were routed with short traces and kept away from switching nets to preserve the accuracy of the 2.59 V and 6.47 V threshold voltages.
- **Noise immunity:** Decoupling capacitor C1 was placed close to the op-amp supply pins to suppress transient noise that could otherwise cause false triggering near the threshold boundaries.
- **Serviceability:** 2-pin headers were used for the battery and supply inputs, allowing easy field wiring to a battery under test and a bench supply.
- **Manufacturability:** Standard through-hole footprints and a two-copper-layer stack-up were used throughout, keeping the design compatible with low-cost PCB fabrication and hand assembly.
- **Verification:** A zero-error ERC, zero-error Ngspice simulation, and a zero-violation DRC with 20/20 connections complete confirm the design is electrically and geometrically sound.

8.2 Conclusion

This task successfully demonstrates the complete PCB design workflow in eSim — from schematic capture in Eeschema, through Ngspice simulation, footprint assignment, PCB layout generation via Sch-to-PCB update, interactive routing, Design Rule Checking, and 3D visualization. The resulting battery over/under-voltage window comparator PCB is fully routed with 20/20 connections complete and zero DRC violations, validating both the correctness of the schematic and the manufacturability of the final board layout. Manufacturing outputs (Gerber and Excellon drill files) were generated for fabrication. The project confirms proficiency with the eSim (KiCad-based) PCB design environment and good practice such as accurate reference/sense network routing, decoupling placement, and clean ERC/DRC verification, meeting the guidelines of the FOSSEE eSim Semester-Long Internship Task 7.